The intensity histogram

The distribution of intensities in each condition, drawn as one curve per condition on a shared axis.

AT A GLANCE

Where	Quality control → Histogram
Data levels	Protein, phosphosite, phosphoprotein
Needs	An uploaded dataset and its annotation. The conditions, and which samples belong to each, come from the annotation.
Data source	The data as loaded, on the $\log_2$ scale. There is no selector: quality control describes what the instrument produced, before filtering or imputation.
One curve	One condition. The value drawn for a feature is its mean across the replicates of that condition.
Table	None.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

1 What the plot answers

Every quantitative comparison in the tool assumes the conditions sit on the same intensity scale. This page is where that assumption is checked: the curves are drawn on one axis, so a condition that is shifted, broader or bimodal is visible against the others rather than in isolation.

Compare both shape and position. A second mode, shoulder or tail can reflect sample composition, processing, technical effects or biology. There is no requirement that a valid pro-

teomics distribution be a single symmetric hump, and this across-feature distribution is not the normality assumption of a feature-wise t-test.

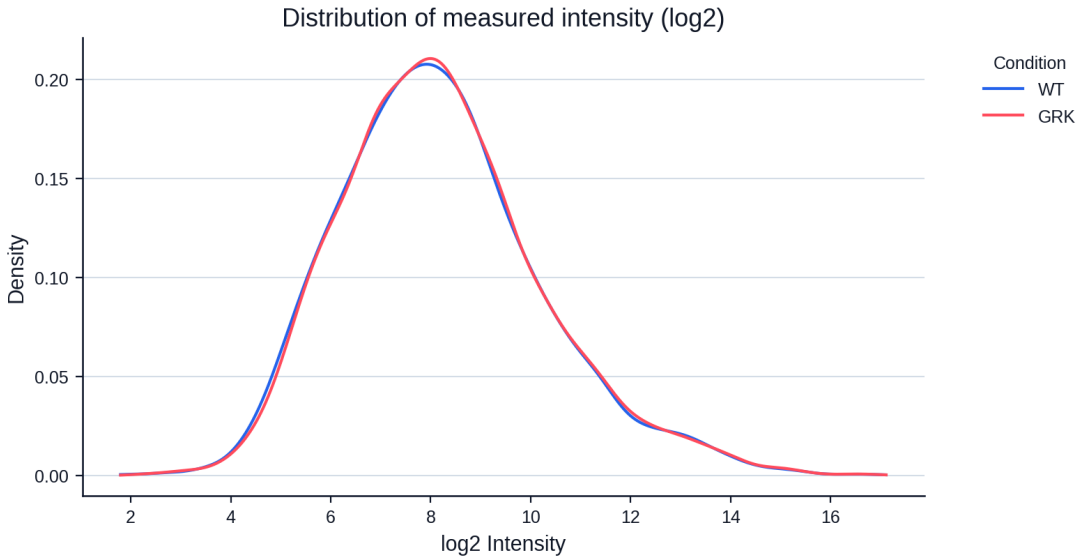


Figure 1.1. Protein level on the example dataset: one curve for the five WT samples and one for the five GRK samples, on the shared $\log_2$ intensity axis. The two lie almost exactly on top of one another — a single hump peaking near 8 with a long right shoulder — which is what two conditions on the same scale look like.

Reading it

- One curve, one condition. Its colour is the condition colour shared with every other plot in the session.
- The x axis: $\log_2$ intensity. At phosphoprotein level collapsed with `robust_site_median` it reads `log2 Ratio to Site Median` instead, because those values are ratios and not abundances.
- The y axis: density. Each curve integrates to one, so a condition built from fewer features is not drawn shorter — the heights are comparable, the counts behind them are not.
- The overlap. Curves that lie on top of one another share a scale; a curve displaced sideways from the rest does not.

> **NOTE**
>
> The curve is a kernel density estimate, not a bar histogram. It is smooth everywhere, including where no feature lies, so a thin tail is the estimator spreading a handful of points rather than evidence of features at those intensities. A condition with fewer than two values, or with no spread at all, cannot be smoothed and is drawn as a 20-bin step histogram instead.

> **EASY TO GET WRONG**
>
> A displaced curve is a difference in how much material was measured at least as often

as it is a difference in biology. Injection amount, digestion efficiency and instrument response all move a whole condition sideways, and none of them is a result. Read the shift here against the boxplot, which shows the same distributions per sample, and against the coverage plot: a condition that is both lower and thinner has a loading problem, while one that is lower but equally wide has been scaled.

2 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Which dataset to draw: protein, phosphosite or phosphoprotein. Only the levels actually uploaded are offered; phosphoprotein appears once a collapse has been run. The selector sits at the top of the page and is shared by every quality-control tab.	Protein	page
Width (cm), Height (cm), DPI	The size of the *downloaded* figure. The browser scales the rendered figure to fit the page. Width and height are held between 1 and 60 cm, DPI between 72 and 600, and a very large figure has its resolution reduced so the image stays under 24 megapixels.	20 cm × 10 cm, 300	page
Download format	PNG, JPG, WEBP, SVG or PDF.	PNG	page

In the plot appearance panel

Control	What it does	Default	Where
Title	Shown or hidden, and its text, size, weight and colour. The title is written by the plot: Distribution of measured intensity (log2).	shown	appearance
Legend	Shown or hidden, and its title, size, colour, frame and position. Its entries are the conditions.	shown	appearance
Condition names, Condition colors	One display name and one colour per condition, for the figure only. The colours are shared with every other plot in the session.	the condition palette	appearance

> **WORTH KNOWING**
>
> The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here.

3 How it is computed

> **HOW IT IS COMPUTED**
>
> The matrix is the data as loaded, on the $\log_2$ scale. Non-numeric and non-finite entries are treated as NA.
> For every feature i and every condition c, the value drawn is the mean over the samples of that condition that carry a value:
>
> $$\bar{x}_{i,c} = \frac{1}{|O_{i,c}|} \sum_{j \in O_{i,c}} x_{ij}, \qquad O_{i,c} = \{\, j \in S_c \,:\, x_{ij} \neq \mathrm{NA} \,\}.$$
>
> A feature missing in every sample of a condition contributes nothing to that condition's curve and is dropped from it.
> The resulting vector is smoothed with a Gaussian kernel density estimate and evaluated on 1000 equally spaced points spanning the smallest to the largest value over *all* conditions, so every curve is drawn on the same grid and the shapes can be compared directly.

> **WHEN THE ESTIMATE CANNOT BE FITTED**
>
> A condition with fewer than two finite means, or whose means are all equal, has no bandwidth to smooth with. That condition alone is drawn as a 20-bin outline histogram, normalised to a density so that it shares the axis with the curves beside it.

Edge cases

- A condition with one replicate is drawn from that sample's own values; the mean over one observation is the observation.
- A feature missing in some replicates of a condition is averaged over the ones it has, so curves can rest on different numbers of features. Nothing on the figure says how many; the coverage plot does.
- If every value in the dataset is identical, the shared axis would have zero width, so it is

widened by one unit either side.

- With no numeric value anywhere in the matrix no figure is drawn and the page reports that instead.

4 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

```python
fig = histo_int(data, meta)
```

```r
fig <- histo_int(data, meta)
```

The function takes no data-source and no level argument: the level is whichever frame is handed in, and the frame is expected on the $\log_2$ scale, as the page's is. `legend=False` and `header=False` drop the legend and the title, and `plot_colors` takes one colour per condition in the order the annotation gives them.

5 Exporting

The download reproduces the plot at the width, height and resolution set on the page. The browser scales the rendered figure to fit the page. Choose PDF for a figure that will be scaled or edited, PNG for a slide.

Behind this page

Function	What it produces
histo_int	One kernel density curve per condition, from the condition-wise mean $\log_2$ intensities